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GOTTA Prototype: Known-Asteroid Extraction and Statistical Performance Evaluation

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Abstract The GOTTA Prototype is a wide-field, high-cadence pathfinder for the full Global Open Transient Telescope Array. Its repeated survey imaging naturally records large numbers of Solar System small bodies, providing an opportunity to validate moving-object processing before the full array is deployed. We present a known-object association workflow for recovering cataloged asteroids from GOTTA Prototype photometric catalogs. For each exposure, the workflow determines the CCD footprint and mid-exposure epoch, predicts known-object ephemerides, filters candidates within the detector footprint, associates predicted positions with calibrated catalog detections, rejects likely stationary-source contaminants using a Gaia cross-check, and augments the accepted matches with orbital and observing-geometry information from external small-body databases. The resulting recovered sample contains 56,230 detections of 16,053 unique known asteroids over 73 UTC nights, including 77 near-Earth objects and 19 potentially hazardous asteroids. The sample is dominated by main-belt asteroids. The observed-minus-predicted positional residual has a median separation of 0.220 arcsec and a two-dimensional RMS of 0.355 arcsec. We characterize the recovered sample in sky position, orbital class, aperture photometry, astrometric residual, angular rate, observing geometry, and repeated temporal sampling. As a pilot light-curve application, we combine selected GOTTA detections with auxiliary 60 cm Schmidt observations and retain 24 period-analysis candidates, including 12 reliable and 12 questionable final classifications in the validation table. These results demonstrate that the GOTTA Prototype has established a working known-object recovery layer and provide a practical baseline for future moving-object association, auxiliary light-curve follow-up, NEO follow-up support, and Solar System studies with the full GOTTA array.

Key words: surveys — minor planets, asteroids: general — techniques: photometric — techniques: astrometric — methods: data analysis

1 INTRODUCTION

Near-Earth objects (NEOs) and potentially hazardous asteroids (PHAs) have long motivated systematic searches for small bodies in the Solar System. The Spaceguard Survey emphasized that the first step toward mitigating impact hazards is a comprehensive search for Earth-crossing asteroids and comets, followed by reliable orbit determination and long-term tracking (Morrison 1992). During the past three decades, wide-field CCD surveys have transformed this task from targeted searches into survey-scale operations. Programs such as Spacewatch, LINEAR, the Catalina Sky Survey, Pan-STARRS, and ATLAS have greatly increased the inventory of known minor planets and have established the basic observing and processing framework for modern moving-object surveys (e.g. Stokes et al. 2000; Larsen 2007; Larson et al. 2003; Denneau et al. 2013; Tonry et al. 2018).

Moving-object science in time-domain surveys depends not only on single-image depth, but also on cadence, sky coverage, astrometric precision, and the ability to associate detections across time. A discovery-oriented pipeline generally starts from transient or non-stationary detections, forms same-night tracklets, and links tracklets obtained on different nights to determine preliminary orbits (Kubica et al. 2007; Denneau et al. 2013; Vereš & Chesley 2017). The same data stream also supports attribution and precovery of known objects, which are essential for orbit refinement, survey validation, and physical characterization. This distinction is particularly important for prototype surveys: even before a full discovery pipeline is finalized, the recovery of cataloged asteroids can be used to evaluate astrometric performance, photometric behavior, source extraction quality, and temporal sampling under real observing conditions.

Several recent and forthcoming surveys illustrate the range of strategies relevant to Solar System time-domain work. ATLAS was designed as a high-cadence all-sky system optimized for the discovery of dangerous near-Earth asteroids and other rapidly varying sources (Tonry et al. 2018). ZTF combines a 47 deg^2 camera with near-real-time reduction and alert distribution for transients, variables, and moving objects (Bellm et al. 2019; Patterson et al. 2019). The Rubin Observatory Legacy Survey of Space and Time will provide much deeper multi-epoch imaging and is expected to substantially expand the number of Solar System detections (Ivezić et al. 2019; Jones et al. 2018;

Schwamb et al. 2023). In parallel, the NEO Surveyor mission will search in the infrared from space and is optimized for discovering and characterizing NEOs, especially objects difficult to observe from the ground (Mainzer et al. 2023). These facilities show that cadence, depth, wavelength, and sky coverage are complementary rather than interchangeable survey properties.

High-cadence survey data are also valuable for asteroid physical characterization. Time-series photometry constrains rotation periods and amplitudes, and multi-geometry light curves can be used to infer spin states and shapes through light-curve inversion (Kaasalainen & Torppa 2001; Kaasalainen et al. 2001; Warner et al. 2009). Sparse survey photometry further enlarges the statistical sample of asteroid light curves when sufficient temporal and phase-angle coverage is available. Accurate photometry and observing geometry are therefore necessary not only for detection and orbit association, but also for studying asteroid rotation, shape, surface properties, and activity. Phase-angle effects must also be treated explicitly when survey photometry is interpreted physically, for example through the H, G_1, G_2 phase-function formalism (Muinonen et al. 2010).

The SiTian/GOTTA program is designed to provide high-cadence, wide-field optical monitoring with a distributed array of telescopes (Liu et al. 2021). Its pathfinder, the Mini-SiTian Array at Xinglong Observatory, has already demonstrated the scientific value of high-cadence, multi-band observations for time-domain astronomy and has provided operational experience for the full SiTian/GOTTA system (Huang et al. 2025; Han et al. 2025). The Mini-SiTian Array consists of three 30 cm telescopes, has operated at Xinglong since 2022, and provides large-field photometric monitoring that informs both observation strategy and data processing for the full SiTian/GOTTA system. These pathfinder data and procedures are directly relevant to the development of Solar System analysis tools for future high-cadence survey operations.

For asteroid science specifically, Liu et al. (2025) developed an asteroid photometry and period-analysis pipeline using Mini-SiTian data from the f02 region. They derived rotation periods for 22 asteroids, including 14 objects without periods listed in ALCDEF, and found agreement with existing results for several objects with sufficiently dense photometric sampling. This work demonstrates the feasibility of asteroid photometry with SiTian pathfinder observations and provides an important motivation for building the corresponding known-object association and validation layer for the GOTTA Prototype.

The GOTTA Prototype extends this pathfinder context to a larger-aperture wide-field system. Mini-SiTian has shown that small pathfinder facilities can deliver useful asteroid light curves, while the GOTTA Prototype provides a testbed for survey-scale source catalogs, exposure geometry, and small-body database augmentation. The present work does not attempt to report unknown asteroid discoveries, measure formal completeness, or operate a full moving-object linking system. Instead, it focuses on known-object recovery, a necessary precursor to tracklet construction, moving-object alert filtering, orbit refinement, and light-curve preparation.

In this paper we present a known-asteroid extraction pipeline for GOTTA Prototype photometric catalogs. We use cataloged minor planets as a controlled sample to validate the end-to-end association between survey catalogs and known-object ephemerides. This approach allows us to evaluate astrometric residuals, photometric measurement behavior, angular-rate coverage, orbital-class distribution, and repeated-observation statistics. The recovered sample contains 56,230 matched detections of 16,053 unique known asteroids, including a small but useful subset of NEOs and PHAs. We also include a pilot light-curve analysis using auxiliary 60 cm Schmidt observations to demonstrate how the recovered known-object catalog can feed into an asteroid period-search workflow.

The paper is organized as follows. Section 2 describes the GOTTA Prototype data and the recovered known-asteroid sample. Section 3 presents the ephemeris-based extraction and augmentation method. Section 4 analyzes the recovered sample and evaluates the association performance. Section 5 describes the light-curve analysis framework and the role of auxiliary observations, and Section 6 discusses the implications for future GOTTA moving-object processing and summarizes the conclusions.

2 OBSERVATIONS AND CATALOG DATA

2.1 GOTTA Prototype Survey Data

The observations analyzed in this work were obtained with the GOTTA Prototype during the original four-CMOS test campaign from 2025 January to June. The prototype is a 1 m-class wide-field Ring Schmidt system located at Xinglong Observatory and was operated during this stage with a single SDSS-like g filter. The optical design, detector layout, and observing configuration are described in detail by Zhengyang Li et al. (in preparation); here we only summarize the aspects required for known-asteroid recovery.

The present analysis uses calibrated source catalogs produced by the GOTTA Prototype basic data-reduction pipeline. That pipeline performs calibration-frame construction, image preprocessing, astrometric calibration, photometric calibration, and source extraction, as described by Niu Li et al. (in preparation). The astrometric solution is tied to Gaia DR3 (Gaia Collaboration et al. 2023), and the photometric catalog provides source positions, aperture photometry, Kron-like photometry, photometric uncertainties, and quality flags for each exposure. Related CMOS wide-field reduction strategies have also been developed for Mini-SiTian (Xiao et al. 2025). For known-asteroid recovery, the most important quantities are the WCS, the exposure time and observing epoch, the measured source coordinates, and the aperture-photometry measurements.

The companion reduction analysis shows that the calibrated catalogs are sufficiently accurate for sub-arcsecond source association. Stellar astrometric residuals are much smaller than the matching scale used here, while the photometric catalog includes aperture-corrected measurements in a sequence of circular apertures. Asteroid detections are generally fainter than the stars used for calibration and may be affected by motion during the exposure. Their observed-minus-predicted residuals therefore include additional contributions from centroiding uncertainty, ephemeris uncertainty, timing uncertainty, and possible trailing.

2.2 Recovered Known-Asteroid Sample

The recovered sample consists of accepted associations between predicted positions of cataloged asteroids and calibrated GOTTA source detections. Each association records the measured source position and photometry, the predicted ephemeris, the observing epoch, and auxiliary orbital and observing-geometry quantities. The final sample contains 56,230 detections of 16,053 unique known asteroids over 73 UTC nights. It is dominated by main-belt asteroids, with smaller contributions from Mars-crossing asteroids, Jupiter Trojans, NEOs, and PHAs. A subset of objects with repeated detections is passed to the light-curve analysis described in Section 5.

Figure 1 shows representative image cutouts of recovered asteroids. These examples illustrate the type of source detections included in the statistical analysis; all quantitative results below are derived from the full recovered sample.

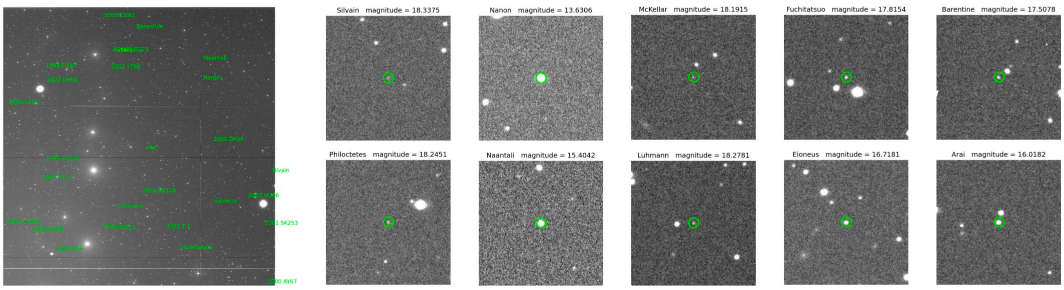

Fig. 1: Representative known-asteroid detections and image cutouts from the pilot-survey material. Green labels and circles identify recovered known asteroids. The figure illustrates the type of image-level detections represented by the accepted associations analyzed in this work.

3 KNOWN-ASTEROID EXTRACTION METHOD

3.1 Known-Object Association Strategy

The goal of the pipeline is to identify detections of cataloged asteroids in calibrated GOTTA Prototype source catalogs. The procedure is designed as a known-object association workflow rather than a discovery-oriented linking pipeline. For each exposure, the pipeline reads the calibrated photometric catalog and the corresponding FITS header. The WCS solution is used to determine the sky footprint of the detector, while the exposure midpoint is computed from the start time and exposure duration. Known-asteroid candidates are predicted for this epoch and for the Xinglong observer location. Objects outside the padded field region, outside the CCD footprint, or fainter than the adopted ephemeris-magnitude limit are removed before cross-matching. The remaining predictions are associated with catalog detections using an angular nearest-neighbor criterion. Likely stationary-source contaminants are rejected with a Gaia DR3 cross-check before the final associations are merged with source photometry and augmented with orbit and observing-geometry quantities from external small-body databases.

Figure 2 summarizes the workflow. The present analysis uses only accepted associations. It is therefore appropriate for evaluating the quality and content of the recovered known-object sample, but not for measuring a formal detection efficiency, which would require the full set of predicted but unrecovered objects.

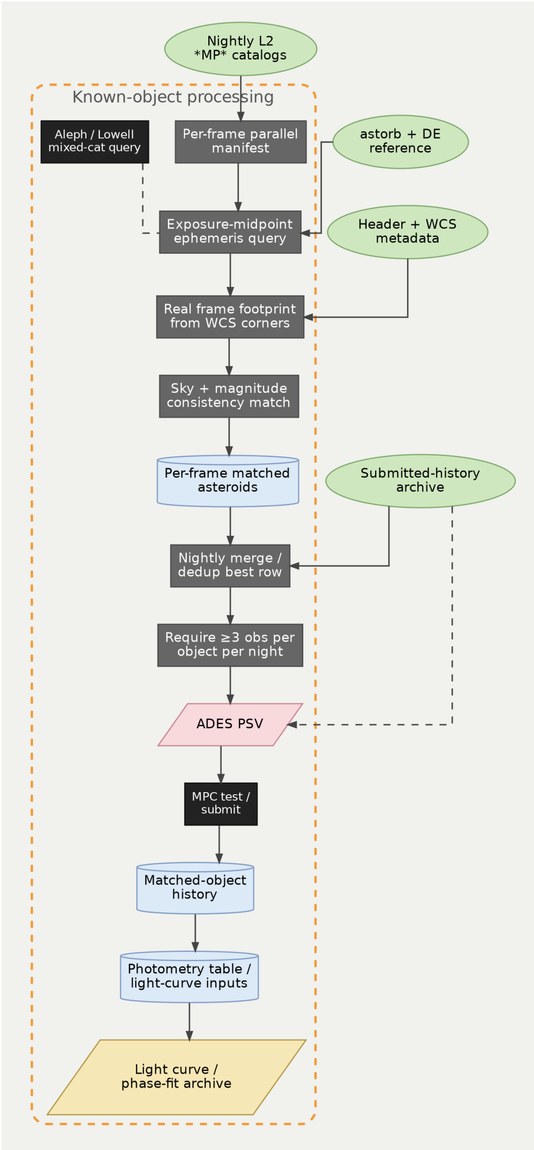

Fig. 2: Workflow of the known-asteroid extraction procedure. The pipeline starts from calibrated per-exposure photometric catalogs and exposure metadata, computes the exposure geometry and observing epoch, predicts known-object positions, applies field and footprint filters, matches surviving predictions to catalog detections, rejects likely stationary Gaia-source contaminants, and augments accepted associations with external small-body information.

3.2 Ephemeris Prediction and Exposure Geometry

For a cataloged asteroid, the starting point of the ephemeris calculation is its osculating orbit at a reference epoch. The orbit can be expressed by the standard elements $(a, e, i, \Omega, \omega, M)$, where a is the semi-major axis, e is the eccentricity, i is the inclination, Ω is the longitude of the ascending node, ω is the argument of perihelion, and M is the mean anomaly. In the two-body limit, the mean anomaly evolves as

$$M(t) = M_0 + n(t - t_0), \tag{1}$$

where n is the mean motion. Kepler's equation,

$$M = E - e \sin E, \tag{2}$$

is solved for the eccentric anomaly E, from which the heliocentric position is obtained. For survey applications, the apparent topocentric direction also depends on the observer location, time scale, light-time correction, and perturbations from Solar System bodies. The pipeline therefore uses a fast ephemeris-screening step to identify plausible in-field candidates and then augments the accepted matches with refined information from the JPL Small-Body Database[1] and Horizons[2] systems. The Minor Planet Center[3] provides the broader astrometric-observation infrastructure and reporting context for Solar System small-body work. The Lowell Observatory astorb database[4] is used as an orbit-catalog resource for fast screening.

The exposure footprint is computed directly from the WCS of each calibrated catalog. The detector corners are projected from pixel coordinates to the sky, and the search radius is initialized from the maximum angular distance between the field center and the projected corners, with an additional padding term to avoid edge losses. The exposure midpoint is

$$t_{\mathrm{mid}} = t_{\mathrm{start}} + \frac{1}{2} t_{\mathrm{exp}}. \tag{3}$$

This midpoint is used for ephemeris evaluation because even modest sky-plane motion can lead to measurable offsets if the start time is used instead. After the initial sky-coordinate query, predicted positions are transformed back to detector coordinates, and only candidates falling within the valid pixel range are retained.

3.3 Catalog Matching and Small-Body Augmentation

The filtered predictions are associated with the GOTTA source catalog using their angular separation from detected sources. The residual between a measured source position and a predicted ephemeris is defined as

$$\Delta\theta = \left[(\Delta\alpha \cos\delta)^2 + (\Delta\delta)^2 \right]^{1/2}, \tag{4}$$

where $\Delta\alpha$ and $\Delta\delta$ are the right ascension and declination differences. A match is accepted only when the nearest-neighbor separation is below the adopted arcsecond-level threshold. The threshold is chosen to be larger than the expected astrometric and centroiding uncertainties for faint moving sources, but sufficiently small to reduce chance associations in crowded fields.

A further verification step is applied to reduce contamination from stationary stellar sources. Because the GOTTA catalog is generated by ordinary source extraction, a predicted asteroid position may occasionally fall close to a field star, and the nearest-neighbor association would then

[1] https://ssd.jpl.nasa.gov/tools/sbdb_lookup.html

[2] https://ssd.jpl.nasa.gov/horizons/

[3] https://minorplanetcenter.net/

[4] https://asteroid.lowell.edu/main/astorb/

select the stellar detection rather than the moving object. To identify such cases, the matched source positions are cross-matched with Gaia DR3 (Gaia Collaboration et al. 2023) after the asteroid-source association step. Associations for which the measured source has a unique Gaia counterpart within 1.5 arcsec and the local GOTTA catalog contains no second source within 1.5 arcsec that could be associated with the moving object are flagged as likely stationary-source contamination and are removed from the final residual analysis. This conservative filter suppresses chance stellar associations before the observed-minus-predicted residuals are examined as a function of magnitude and sky-plane angular rate.

For each accepted association, the measured catalog quantities are merged with the predicted ephemeris. The photometric measurements include aperture magnitudes and their uncertainties; Kron-like measurements are retained for comparison but are not used as the primary photometric statistic in the final analysis. The predicted asteroid magnitude is used only as a consistency diagnostic, because it is not necessarily defined in the same photometric system as the GOTTA *g*-band measurement and is also affected by asteroid color, phase behavior, and rotational variability.

The matched detections are further associated with small-body database entries to obtain object identifiers, orbit classes, orbital elements, and observer-dependent geometry. The latter includes heliocentric distance, topocentric distance, phase angle, and sky-plane angular rates. These quantities are used in the statistical analysis below to interpret the recovered sample in terms of orbital population, observing geometry, and apparent motion.

4 RESULTS

4.1 Recovered Sample and Orbital Populations

The recovery procedure yields 56,230 accepted associations corresponding to 16,053 distinct known asteroids. These detections span 73 UTC nights during the 2025 January–June prototype campaign. The size of this sample is sufficient to characterize the dominant orbital populations recovered by the survey and to examine how the association residuals depend on brightness and apparent motion. Table 1 summarizes the main statistics.

The orbit-class composition is shown in Table 2. Main-belt asteroids account for 51,058 detections and 14,674 unique objects. Outer main-belt asteroids, Jupiter Trojans, inner main-belt asteroids, and Mars-crossing asteroids make up most of the remaining sample. The NEO classes are represented by Amor, Apollo, and Aten objects. The dominance of main-belt objects is expected because the survey footprint and limiting depth preferentially sample objects near ordinary main-belt opposition geometries. The smaller NEO and PHA subsets should therefore be viewed as validation samples for the recovery path rather than as indicators of discovery performance.

The recovery rate also varies substantially from night to night because of sky coverage, observing conditions, and the number of available exposure catalogs. The five most productive UTC nights are listed in Table 3.

The recovered detections are concentrated in the surveyed sky regions available during the prototype campaign, reflecting the combination of the survey footprint, seasonal visibility, and

Table 1: Statistics for the recovered GOTTA known-asteroid sample.

Quantity	Value	Note
Accepted source-ephemeris associations	56,230	Accepted source-ephemeris associations
Distinct known asteroids	16,053	Unique minor-planet identifiers
Exposure catalogs with accepted associations	13,253	Per-exposure catalog products
UTC nights represented	73	Recovered sample date coverage
Field groups represented	588	Grouped by pointing metadata
NEO-class objects	77	Identified from small-body metadata
PHA-class objects	19	Identified from small-body metadata
Median ephemeris magnitude	18.2215	Predicted brightness
Median g_{aper}	18.1110	Adopted aperture magnitude
Median $\sigma(g_{\mathrm{aper}})$	0.1239 mag	Adopted aperture uncertainty
Median aperture S/N proxy	8.7621	Flux/error in adopted aperture
Median observed-minus-predicted separation	0.220 arcsec	Recovered detections
84th-percentile observed-minus-predicted separation	0.484 arcsec	Recovered detections
2D residual RMS	0.355 arcsec	sqrt(mean(sep^2))
RMS in $\Delta\alpha\cos\delta$	0.250 arcsec	Coordinate residual
RMS in $\Delta\delta$	0.252 arcsec	Coordinate residual
Median angular rate	29.276 arcsec hr^{-1}	Derived from augmented geometry
Median phase angle	13.245 deg	Derived from augmented geometry
Associations without geometry augmentation	107	Excluded from geometry statistics

Table 2: Orbit-class composition of the recovered known-asteroid sample. Separations are in arcsec.

Code	Orbit class	Detections	Unique objects	Median g_{aper}	Median separation
MBA	Main-belt Asteroid	51,058	14,674	18.123	0.221
OMB	Outer Main-belt Asteroid	1,970	522	17.893	0.187
TJN	Jupiter Trojan	1,362	317	18.005	0.207
IMB	Inner Main-belt Asteroid	897	290	18.315	0.257
MCA	Mars-crossing Asteroid	656	166	17.847	0.228
AMO	Amor	139	31	17.143	0.227
APO	Apollo	101	38	17.749	0.253
TNO	TransNeptunian Object	20	3	17.588	0.243
ATE	Aten	19	8	17.668	0.292
CEN	Centaur	5	3	18.481	0.237
AST	Asteroid	3	1	17.074	0.199

the instantaneous distribution of known minor planets. The orbital distribution confirms that the sample is dominated by main-belt objects, with smaller contributions from inner and outer

Table 3: Five UTC nights with the largest numbers of accepted known-asteroid associations. These counts reflect the combination of observing coverage, sky position, and catalog availability on each night.

UTC date	Detections	Unique objects	Exposure catalogs	Median g_{aper}	Median separation
2025-02-05	3,385	3,362	353	18.739	0.187
2025-01-19	3,119	3,065	576	18.354	0.189
2025-03-05	2,657	2,639	372	18.439	0.220
2025-01-28	2,656	2,638	395	18.499	0.189
2025-01-29	2,645	2,621	307	18.498	0.201

main-belt asteroids, Jupiter Trojans, Mars-crossers, and NEOs. These distributions are shown in Figures 3 and 4.

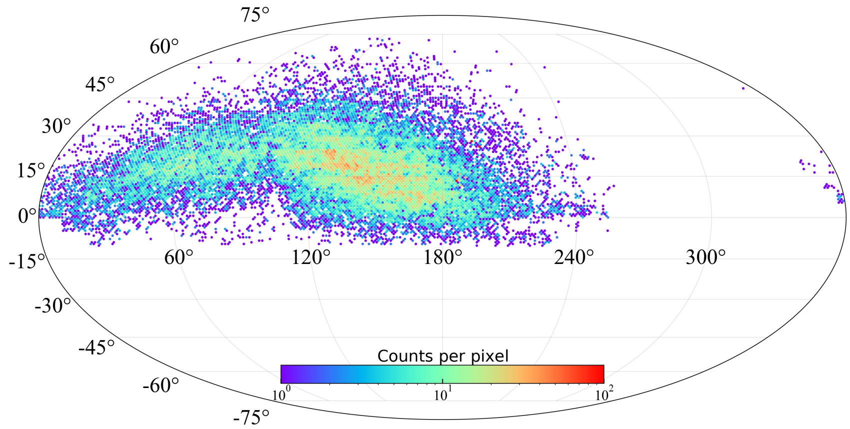

Fig. 3: Detection-level sky distribution of recovered known asteroids in equatorial coordinates, binned with HEALPix nside 64. The color scale is clipped at 100 detections per pixel for readability.

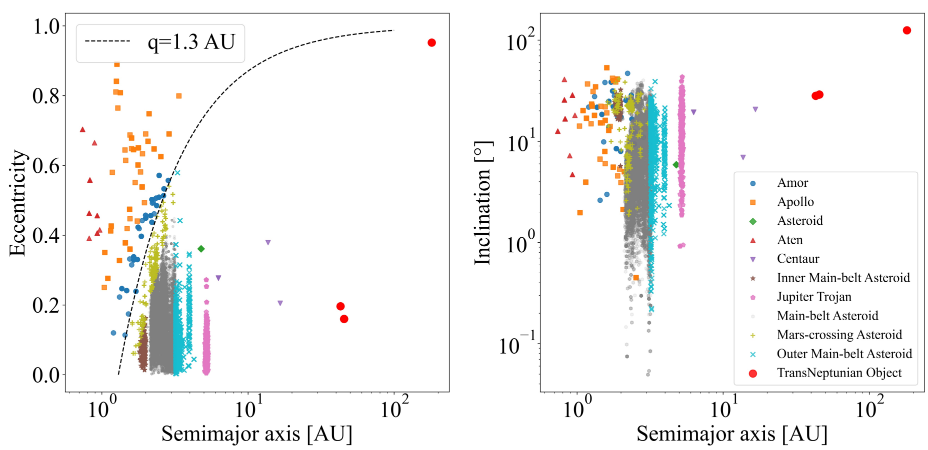

Fig. 4: Orbital-element distribution of the recovered known-asteroid sample. Points are colored by orbit class. The dashed curve in the semi-major-axis–eccentricity panel marks perihelion distance $q = 1.3$ AU, a conventional boundary for near-Earth objects.

Having established that the recovered catalog is dominated by main-belt objects, we next examine whether the measured brightness and photometric uncertainties are consistent with a sample of mostly faint single-exposure asteroid detections.

4.2 Photometric Properties of the Recovered Asteroids

For the photometric analysis we use the aperture magnitude corresponding to the optimal aperture defined by the basic GOTTA photometric pipeline. In that pipeline, aperture photometry is measured in a series of circular apertures, and the optimal aperture is determined from the signal-to-noise ratio of stellar sources as a function of aperture radius. A larger reference aperture is used for aperture correction, so that the cataloged aperture magnitudes are placed on a consistent photometric scale. We therefore do not re-derive an asteroid-specific optimal aperture in this work. Instead, we use the pipeline-defined optimal aperture as a homogeneous measurement for the recovered asteroid sample.

This choice is preferable to Kron-like photometry for the statistical analysis presented here. Kron apertures are adaptive and can respond differently to background noise, faint-source segmentation, and possible source elongation, whereas the adopted aperture magnitude provides a uniform measurement across the sample. Kron photometry is retained in the catalog for comparison, but it is not used as the primary quantity in the magnitude, uncertainty, and astrometric-residual diagnostics below.

We denote the adopted aperture magnitude as g_{aper} below. Most recovered asteroids lie close to the faint end of routine single-exposure catalog measurements rather than in the high signal-to-noise regime used for stellar calibration. The uncertainty–magnitude relation increases toward the faint end, as expected from decreasing source signal-to-noise ratio. The difference between predicted ephemeris magnitude and measured GOTTA g-band aperture magnitude is used only as a consistency diagnostic, because the two quantities are not necessarily defined in the same photometric system and may also differ because of asteroid color, phase behavior, rotation, shape, and ephemeris-magnitude uncertainties. The corresponding photometric distributions are summarized in Figure 5.

Because the photometric uncertainty rises rapidly toward the faint end, the same magnitude dependence is expected to appear in the astrometric residuals through centroiding uncertainty.

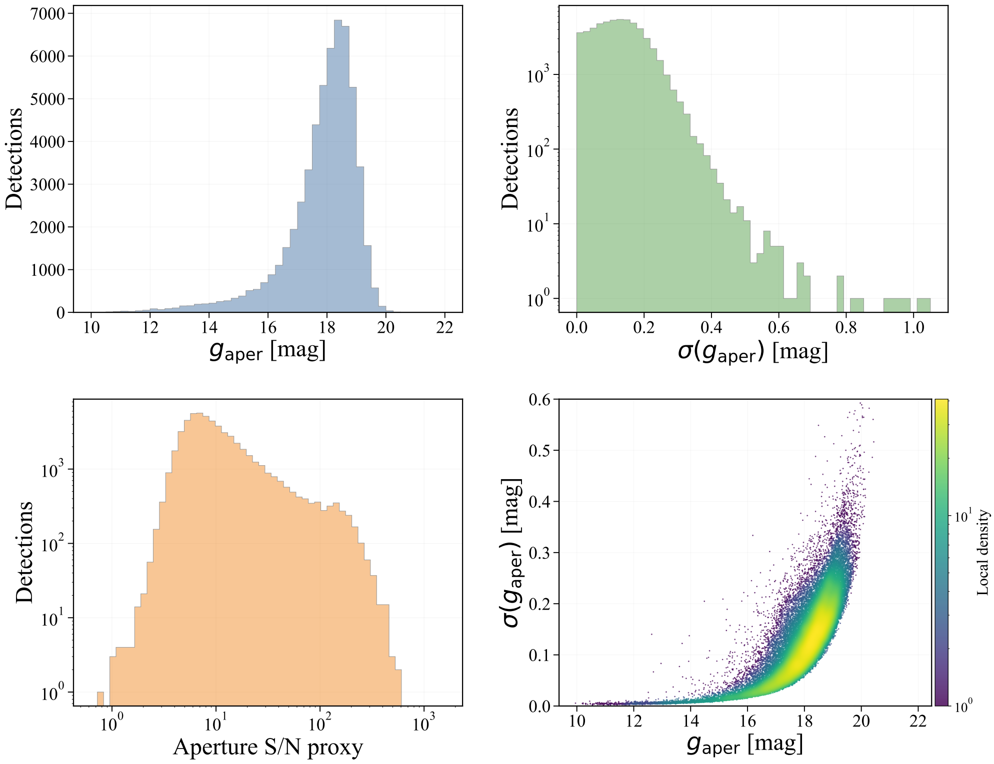

Fig. 5: Aperture-photometry diagnostics for recovered known asteroids. The adopted aperture follows the optimal aperture defined by the GOTTA basic photometric pipeline. The panels show the adopted aperture magnitude distribution, the corresponding magnitude-uncertainty distribution, the aperture S/N proxy, and the dependence of magnitude uncertainty on adopted aperture magnitude. The last panel is shown as a density map to emphasize the dominant distribution of faint asteroid detections.

4.3 Astrometric Residuals and Apparent Motion

The residual distribution is strongly concentrated below the adopted matching scale, with a median separation of 0.220 arcsec and an 84th percentile of 0.484 arcsec. The two-dimensional RMS is 0.355 arcsec, and the RMS values of $\Delta\alpha\cos\delta$ and $\Delta\delta$ are 0.250 arcsec and 0.252 arcsec, respectively. The coordinate residuals are centered close to zero, indicating that the accepted associations do not show a large global offset between the predicted ephemerides and the calibrated source positions. When the residuals are grouped by adopted aperture magnitude, the median separation increases from about 0.09–0.11 arcsec for relatively bright detections to about 0.27 arcsec near $g_{\rm aper} \simeq 18$–19 mag. This trend indicates that source centroiding and signal-to-noise ratio dominate the residual budget for most recovered asteroids.

The dependence on sky-plane angular rate is weaker over the main part of the sample. Most recovered objects have moderate rates, with a median of 29.276 arcsec hr^{-1}. In this regime, the median separation remains close to the full-sample value. The sparsely populated high-rate tail shows larger scatter and should be interpreted with caution because it contains few detections and may be more affected by trailing, centroiding biases, or matching ambiguity.

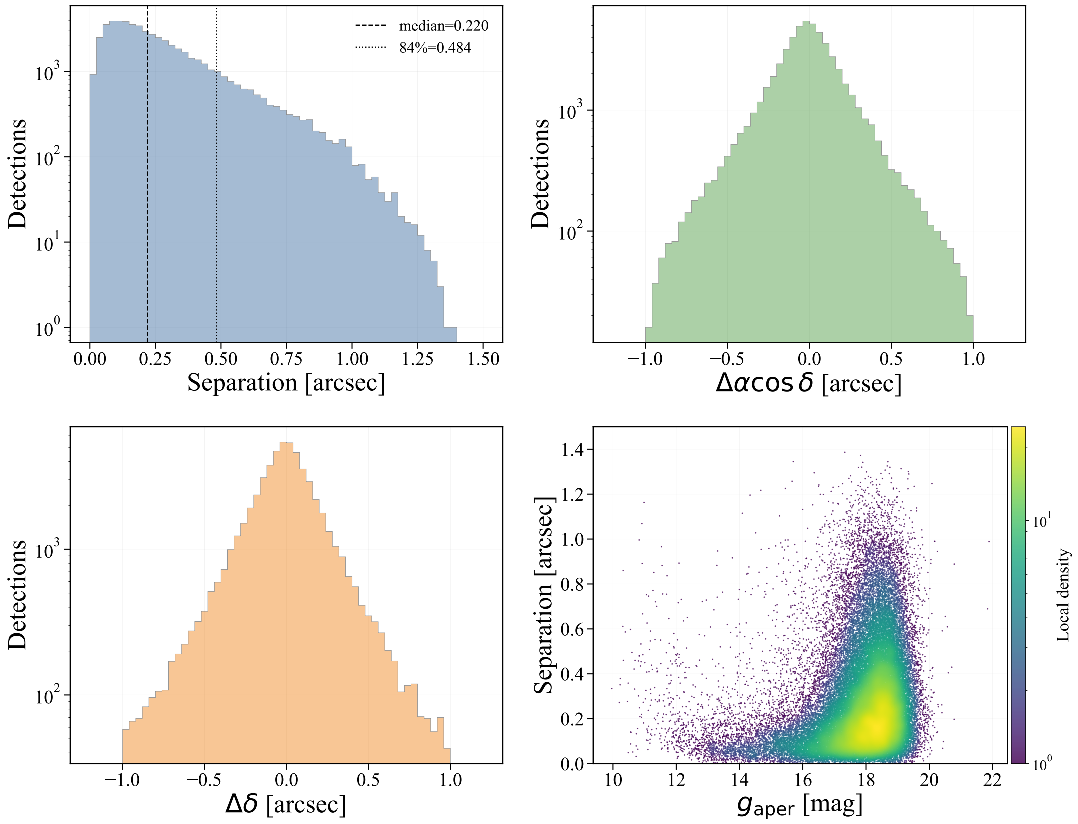

Fig. 6: Astrometric residual diagnostics. The panels show the observed-minus-predicted separation, $\Delta\alpha\cos\delta$, $\Delta\delta$, and separation as a function of the adopted aperture magnitude $g_{\rm aper}$.

Table 4: Astrometric residuals as a function of adopted aperture magnitude $g_{\rm aper}$. Separations are in arcsec.

Magnitude bin	Detections	Median separation	RMS separation	Median mag. error
$[12, 13)$	340	0.113	0.210	0.005
$[13, 14)$	698	0.086	0.174	0.007
$[14, 15)$	1,072	0.089	0.194	0.010
$[15, 16)$	2,129	0.104	0.225	0.018
$[16, 17)$	5,447	0.135	0.259	0.039
$[17, 18)$	15,625	0.204	0.343	0.085
$[18, 19)$	24,990	0.270	0.391	0.155
$[19, 20)$	5,687	0.268	0.380	0.234

The corresponding distributions are shown in Figures 6 and 7, and the binned statistics are summarized in Tables 4 and 5.

An example exposure-level diagnostic is shown in Figure 8. The accepted associations are distributed across the detector, and the residual vectors do not show a coherent direction in this example. This diagnostic is intended to illustrate the matching behavior after the Gaia stationary-source filter; the statistical residual properties are quantified by the full sample rather than by this single exposure.

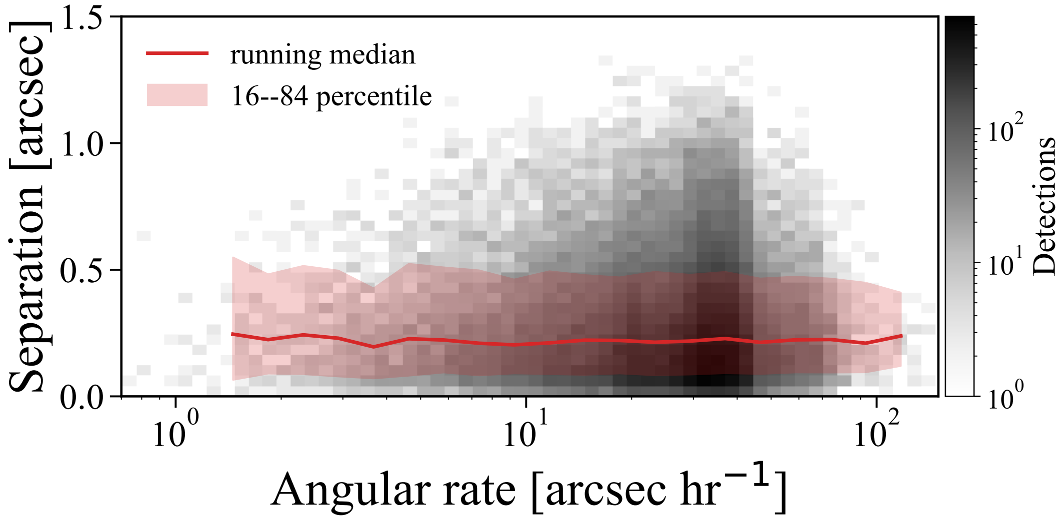

Fig. 7: Observed-minus-predicted separation as a function of sky-plane angular rate. The gray density map shows the dominant range of the recovered sample, while the red curve and shaded region show the running median and the 16th–84th percentile range in logarithmic rate bins. Most detections lie at moderate angular rates, where the residual distribution remains close to the full-sample behavior. A small number of higher-rate detections are excluded from the displayed range for clarity but are included in the binned statistics.

Table 5: Astrometric residuals as a function of sky-plane angular rate. Rate bins are in arcsec hr^{-1} and separations are in arcsec.

Rate bin	Detections	Median separation	RMS separation	Median g_{aper}
$[0, 10)$	4,917	0.213	0.356	18.053
$[10, 20)$	10,841	0.218	0.351	18.083
$[20, 40)$	31,792	0.222	0.357	18.136
$[40, 80)$	8,262	0.219	0.348	18.099
$[80, 160)$	224	0.222	0.337	17.790
$[160, 320)$	64	0.253	0.358	17.699
$[320, 1000)$	20	0.467	0.543	17.389

The weak residual dependence on angular rate indicates that, for most detections, the sample is limited more by brightness and centroiding precision than by apparent motion. We therefore turn next to the observing geometry and revisit statistics that determine the time-domain usefulness of the recovered objects.

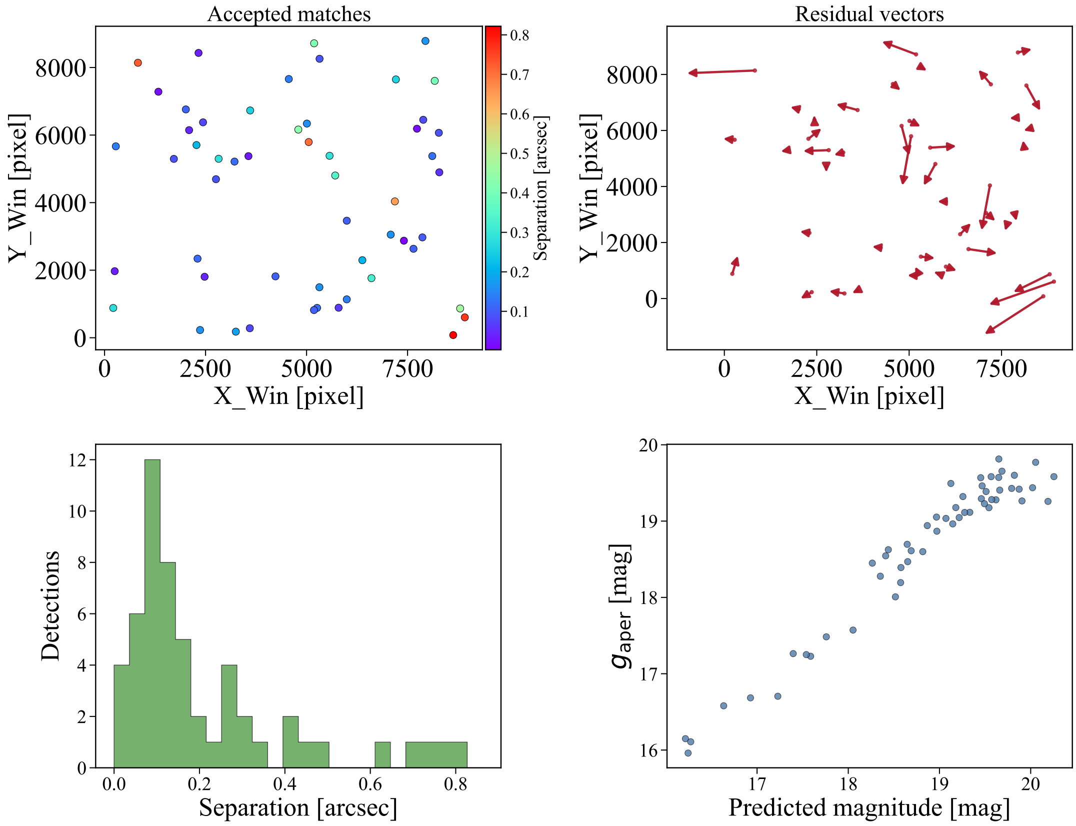

Fig. 8: Example cross-matching diagnostic for one exposure. The panels show the detector distribution of accepted known-asteroid associations, residual vectors between predicted and measured positions, the separation distribution, and the relation between predicted ephemeris magnitude and measured aperture magnitude. The vector lengths are scaled for visibility. Associations flagged as likely stationary Gaia sources are not included.

4.4 Temporal Sampling and Light-Curve Potential

The augmented geometry confirms that the recovered sample is dominated by ordinary main-belt observing configurations. The median heliocentric distance is 2.581 AU, the median topocentric distance is 1.840 AU, and the median phase angle is 13.245 deg. These values explain why the dominant angular-rate distribution is relatively narrow and why most detections are not strongly affected by extreme sky-plane motion.

The scientific value of a high-cadence survey for asteroid work depends not only on the number of recovered objects, but also on how often individual objects are revisited. In the current sample the revisit statistics are highly non-uniform. The median number of detections per object is two, the mean is 3.50, and the 84th percentile is six. A small subset of objects has much denser sampling; the most frequently recovered object has 317 detections over nine distinct nights. These repeatedly recovered objects are useful for time-series construction tests, although they are not representative of the full sample. Table 6 lists the five most frequently recovered objects.

These repeated detections are useful for testing the construction of asteroid time series and for checking internal consistency of the photometry. However, they do not yet imply that the prototype-

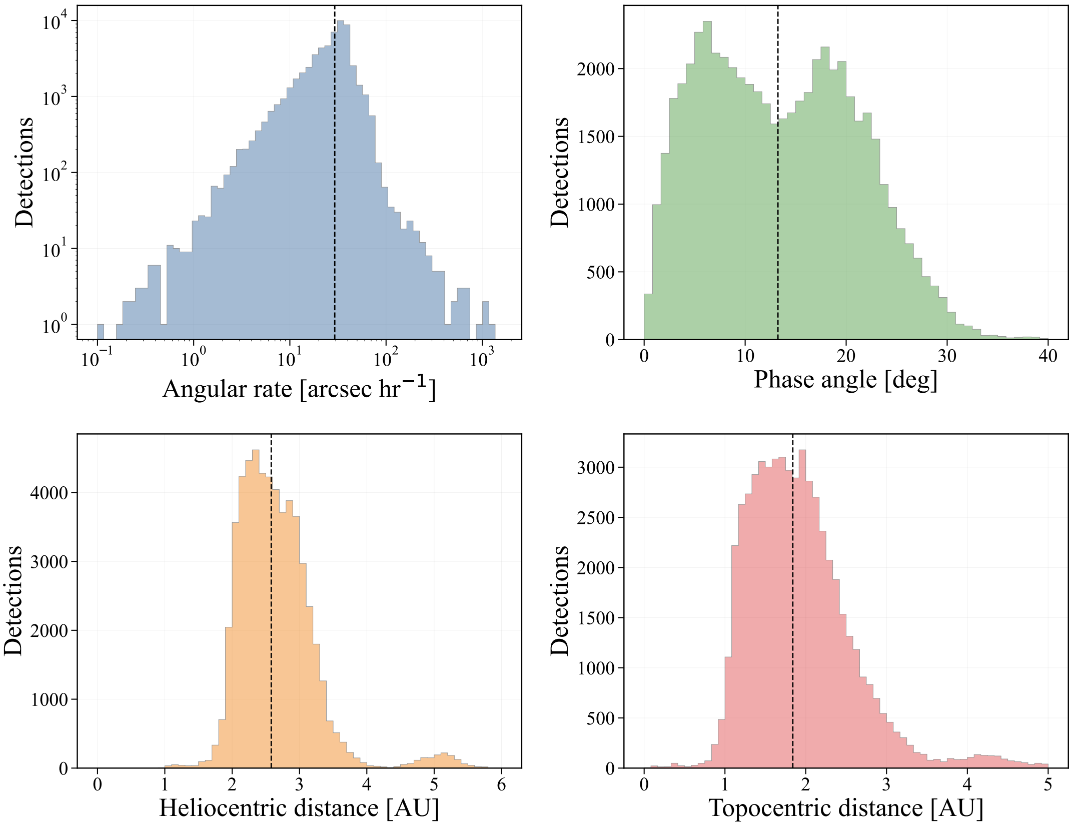

Fig. 9: Observing-geometry diagnostics for recovered known asteroids with successful geometry augmentation. The panels show sky-plane angular rate, phase angle, heliocentric distance, and topocentric distance. Vertical dashed lines mark the corresponding medians.

Table 6: Five most frequently recovered known asteroids in the GOTTA Prototype sample. These objects are useful for time-series construction tests, but they are not representative of the full sample.

Object ID	Name	Detections	Nights	First MJD	Last MJD	Baseline (days)	Median g_{aper}	Class
214386	2005 MU1	317	9	60739.9	60818.7	78.824	18.141	MBA
76719	2000 JJ18	256	3	60738.8	60798.8	59.974	18.595	MBA
293	Brasilia	67	14	60709.9	60809.6	99.701	15.233	MBA
15751	1991 VN4	47	9	60714.8	60809.6	94.840	17.684	MBA
33629	1999 JK76	40	13	60709.9	60818.7	108.732	17.859	MBA

only survey can deliver reliable rotation periods for most recovered asteroids. The cumulative detection-count and baseline distributions show that most objects have sparse sampling and limited phase coverage.

For most objects, however, the prototype-only measurements are too sparse for reliable period determination. This is the motivation for the auxiliary light-curve experiment described in Section 5: the recovered GOTTA asteroid catalog provides the target list and part of the photometric time series, while the nearby 60 cm Schmidt telescope supplies the denser sampling required to test the period-search pipeline.

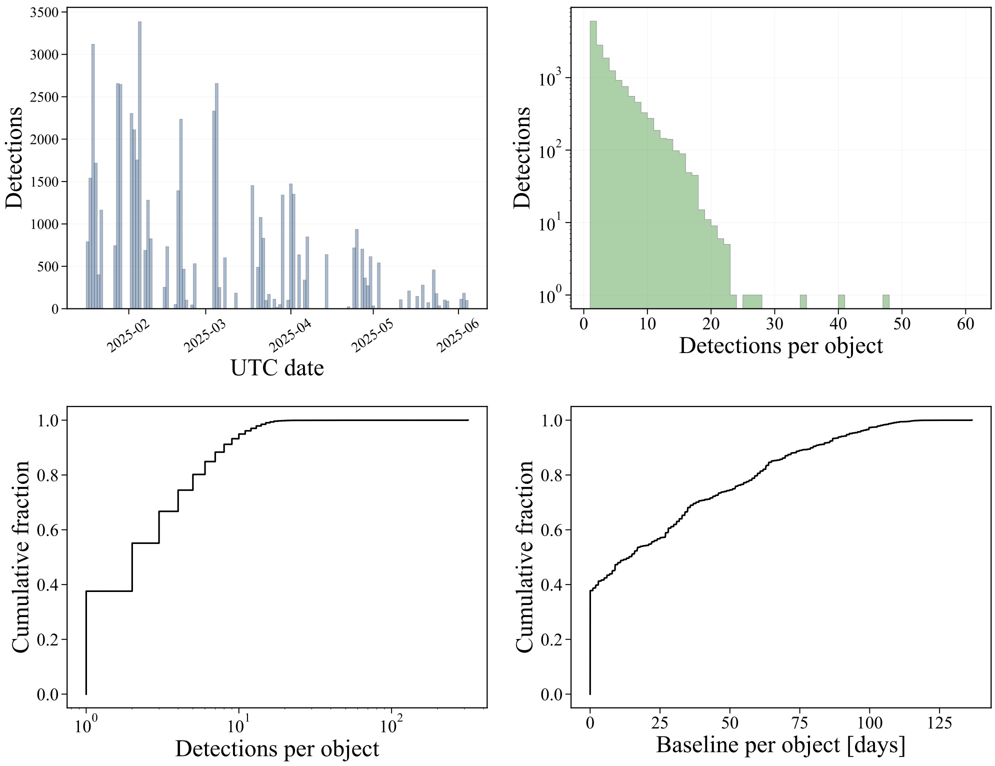

Fig. 10: Temporal sampling of the recovered known-asteroid sample. The panels show the nightly number of accepted associations, the distribution of detections per object, the cumulative detection-count distribution, and the cumulative distribution of object time baselines.

5 LIGHT-CURVE ANALYSIS WITH AUXILIARY OBSERVATIONS

5.1 Motivation and Input Data

The repeated-detection statistics in Section 4 show that the GOTTA Prototype catalog provides useful sparse time-domain information for known asteroids, but that the prototype-only sampling is insufficient for robust rotation-period determination for most recovered objects. The median number of detections per object is only two, and only a small subset of objects has enough repeated measurements to constrain a folded light curve. We therefore use the GOTTA Prototype measurements primarily as an input layer for light-curve construction and combine selected targets with additional observations obtained by a nearby 60 cm Schmidt telescope. The purpose of this combined analysis is to validate the period-search component of the asteroid pipeline under realistic observing conditions, rather than to produce a complete period catalog from the prototype survey alone.

For each target, we construct light curves from the available photometric measurements and apply quality control separately to aperture, PSF, and Kron-like photometry. Measurements with missing values or obvious outliers are removed, and a light curve is retained for period analysis only if it contains at least 60 valid points. This requirement reduces the number of spurious periodogram peaks caused by sparse sampling and provides a minimum phase coverage for model comparison. When multiple photometric measurements are available for the same target, they are treated as independent checks on the candidate period. Only data sets with negligible inter-night or inter-instrument zero-point offsets are used in the current demonstration. For the short time spans considered here, changes in heliocentric distance and topocentric distance are small for most selected targets; the pipeline nevertheless records the observing geometry so that reduced magnitudes or phase corrections can be applied in future analyses.

5.2 Period Search and Model Selection

Candidate periods are searched using three complementary methods. The Lomb–Scargle periodogram is used as the primary method for unevenly sampled time-series data (Lomb 1976; Scargle 1982). Phase Dispersion Minimization (PDM) provides an independent search that is less sensitive to the assumption of a sinusoidal light-curve shape and is therefore useful for asteroid light curves (Stellingwerf 1978; Warner et al. 2009). A frequency-domain diagnostic based on a Fourier transform of an interpolated, uniformly sampled light curve is also used as an auxiliary check. Because the effective time baselines of the current light curves are limited, the search is restricted to short-period solutions, with the adopted candidate periods required to be below 10 hr.

Candidate periods from the different search methods and photometric measurements are grouped in relative-period space. Periods that agree within 10 per cent are assigned to the same cluster, and the median period of the largest cluster is used as the initial candidate period. If no repeated period cluster is found, the object is not assigned a robust period at this stage. This method avoids relying on a single periodogram peak and provides a simple consistency requirement across photometric measurements and search algorithms. We record this agreement through a match-point metric across the Lomb–Scargle, PDM, and FFT searches; the intermediate search-quality flag in the validation table is based on this metric and records the consistency of the period-search candidates before the final folded-light-curve assessment. The initial clustering is performed without using an external reference period; literature periods are consulted only as a later comparison.

Asteroid light curves often show double-peaked morphology because the projected shape generally repeats twice during one rotation. We therefore test both P and $2P$ for each candidate period. For double-peaked light curves, the adopted rotation period is the double-peaked period. The phase is computed as

$$\phi = \left[\frac{t - t_0}{P}\right] \bmod 1, \tag{5}$$

where t_0 is an arbitrary reference epoch. The folded light curve is fitted with a Fourier series,

$$m(\phi) = a_0 + \sum_{k=1}^{N} \left[a_k \cos(2\pi k\phi) + b_k \sin(2\pi k\phi)\right], \tag{6}$$

where $N \leq 3$ in the current implementation. The fit is performed with photometric-error weighting and iterative sigma clipping to reduce the influence of outliers. The model order and the choice between P and $2P$ are evaluated using the residuals and the Bayesian Information Criterion (Schwarz 1978),

$$\mathrm{BIC} = n \ln(\sigma^2) + k \ln n, \tag{7}$$

where n is the number of valid measurements, k is the number of model parameters, and σ is the residual scatter.

To identify plausible double-peaked solutions, we also examine the harmonic content and morphology of the Fourier model. In particular, the relative strength of the second harmonic,

$$R_{21} = \frac{A_2}{A_1}, \tag{8}$$

is used together with the number of peaks, phase coverage, phase-overlap statistics, and the residual improvement between P and $2P$. High-confidence solutions are visually inspected to reject obvious aliases and poorly covered folded light curves. For accepted solutions, the period uncertainty is estimated from a local scan of the fit statistic around the adopted period.

5.3 Candidate Classification and Preliminary Results

The combined GOTTA and 60 cm Schmidt data are used to identify asteroid light curves with plausible rotational modulation. The updated analysis retains 24 objects in the period-analysis validation table. The final quality flag is assigned after inspecting the folded light curve and the sampling ratio between the usable time span and the adopted period, rather than from the period-search agreement metric alone. Twelve objects are assigned reliable final quality flags, while the remaining twelve are retained as questionable solutions that require additional observations for confirmation. None of the 24 candidate periods has a corresponding reference period in the external period database used in the current comparison. This statement should be checked against the final database query before submission and should not be interpreted as a complete claim of newly discovered periods.

Appendix Figures A.1 and A.2 show folded light curves for the 24 objects retained in the updated validation sample, separated by final quality flag so that each mosaic can be displayed at full-page scale. The panel titles list the initial period and the adopted single- or double-peaked period used for the displayed phase folding. The black curves show the Fourier models used for the morphology check. Appendix Table A.1 summarizes the final period-analysis validation table, including the total and effective photometric point counts, the adopted period, period uncertainty, period model, preferred photometric measurement type, search-quality flag, and final-quality flag. The table is sorted by final quality, with reliable solutions first, and then by object identifier within each quality group. These results demonstrate that, although the GOTTA Prototype data alone are generally too sparse for period determination, the recovered known-object catalog can be combined with auxiliary observations to validate the asteroid period-analysis pipeline.

6 DISCUSSION AND CONCLUSIONS

The recovered known-asteroid sample demonstrates that calibrated GOTTA Prototype catalogs can support systematic association with cataloged Solar System objects. The main validation provided by this work is the closure of the known-object association workflow: calibrated source catalogs are combined with exposure geometry and mid-exposure timing, known-object ephemerides are filtered by the real CCD footprint, likely stationary-source contaminants are removed using Gaia cross-checks, accepted detections are matched in the source catalog, and the resulting associations are augmented with small-body database information. This workflow is a prerequisite for future moving-object products, including candidate-level bookkeeping, tracklet construction, and moving-object-aware transient filtering.

Known-object recovery is a useful validation mode because the predicted positions of cataloged asteroids are obtained independently of the GOTTA source extraction. The observed-minus-predicted residuals therefore test the combined effects of WCS quality, timing accuracy, centroiding, ephemeris uncertainty, and matching behavior. The median separation of 0.220 arcsec and the two-dimensional RMS of 0.355 arcsec show that the accepted associations are well localized on the scale required for known-object recovery. The increase of residual with fainter aperture magnitude indicates that centroiding precision and signal-to-noise ratio dominate the residual distribution for most of the sample. The weak dependence on angular rate over the main rate range suggests that trailing is not the primary limitation for the bulk of recovered objects, although the sparsely populated high-rate tail still requires dedicated analysis.

The photometric results should be interpreted as survey-catalog diagnostics rather than asteroid physical measurements. The adopted aperture magnitude provides a homogeneous brightness estimate, but the ephemeris magnitude is not a calibrated reference in the GOTTA g band. Asteroid color, rotational modulation, phase-angle dependence, shape, and uncertainties in the magnitude model can all contribute to the difference between predicted and observed brightness. The light-curve analysis in Section 5 illustrates how the recovered known-object catalog can be used as the input layer for asteroid time-series work. The GOTTA-only sampling is generally too sparse for period determination, but the addition of 60 cm Schmidt observations allows the period-search machinery to be tested on selected targets. The reliable periods obtained from the combined data should therefore be interpreted as a pipeline validation sample rather than as a complete rotational census.

Several limitations should be kept in mind when interpreting the present recovered sample. Because the catalog contains only accepted associations, it does not provide the denominator required for a formal recovery-efficiency estimate. The Gaia cross-check removes obvious stationary-source contaminants, but it does not by itself provide a complete false-association model. A quantitative false-match rate still requires the full set of predicted candidates, rejected associations, and original per-exposure source catalogs, together with random-shift or time-scrambling tests. A quantitative trailing analysis will also require exposure-level seeing and exposure-time information, so that the expected trail length,

$$L_{\mathrm{trail}} = \mu t_{\mathrm{exp}}, \tag{9}$$

can be compared with astrometric residuals and photometric uncertainties. Here μ is the sky-plane angular rate and t_{exp} is the exposure time.

For the future GOTTA array, the strength of the system will lie in the combination of large sky coverage and frequent temporal sampling. The prototype results show that the association and augmentation framework needed for known-object recovery can be implemented on GOTTA data. For the full GOTTA array, denser revisit patterns and multi-site coverage should increase the number of objects with sufficient phase coverage for period recovery, reducing the need for auxiliary observations for many targets. With candidate-level bookkeeping, repeated field visits, multi-band observations, and moving-object-aware scheduling, the same framework can be extended toward NEO monitoring, asteroid light-curve studies, and eventually discovery-oriented moving-object processing.

In summary, we have developed and tested a known-asteroid recovery workflow for GOTTA Prototype photometric catalogs. The recovered sample contains 56,230 detections of 16,053 unique known asteroids over 73 UTC nights, including small subsets of NEOs and PHAs. The sample is dominated by main-belt asteroids and provides statistically useful distributions in sky position, orbital class, aperture magnitude, angular rate, observing geometry, astrometric residual, and repeated temporal sampling. The sub-arcsecond observed-minus-predicted residuals demonstrate that the current catalog products are suitable for known-object association. The pilot light-curve analysis retains 24 period-analysis candidates, with 12 reliable and 12 questionable final classifications in the combined GOTTA and 60 cm Schmidt validation table. This work establishes the first moving-object validation layer for the GOTTA Prototype and provides the basis for future candidate-level detection, tracklet construction, and asteroid light-curve analysis.

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Software: Astropy (Astropy Collaboration et al. 2013, 2018, 2022), NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), scikit-learn (Pedregosa et al. 2011), Matplotlib (Hunter 2007), pandas (McKinney 2010), astroquery (Ginsburg et al. 2019), and healpy/HEALPix (Górski et al. 2005).

Appendix A: LIGHT-CURVE VALIDATION PRODUCTS

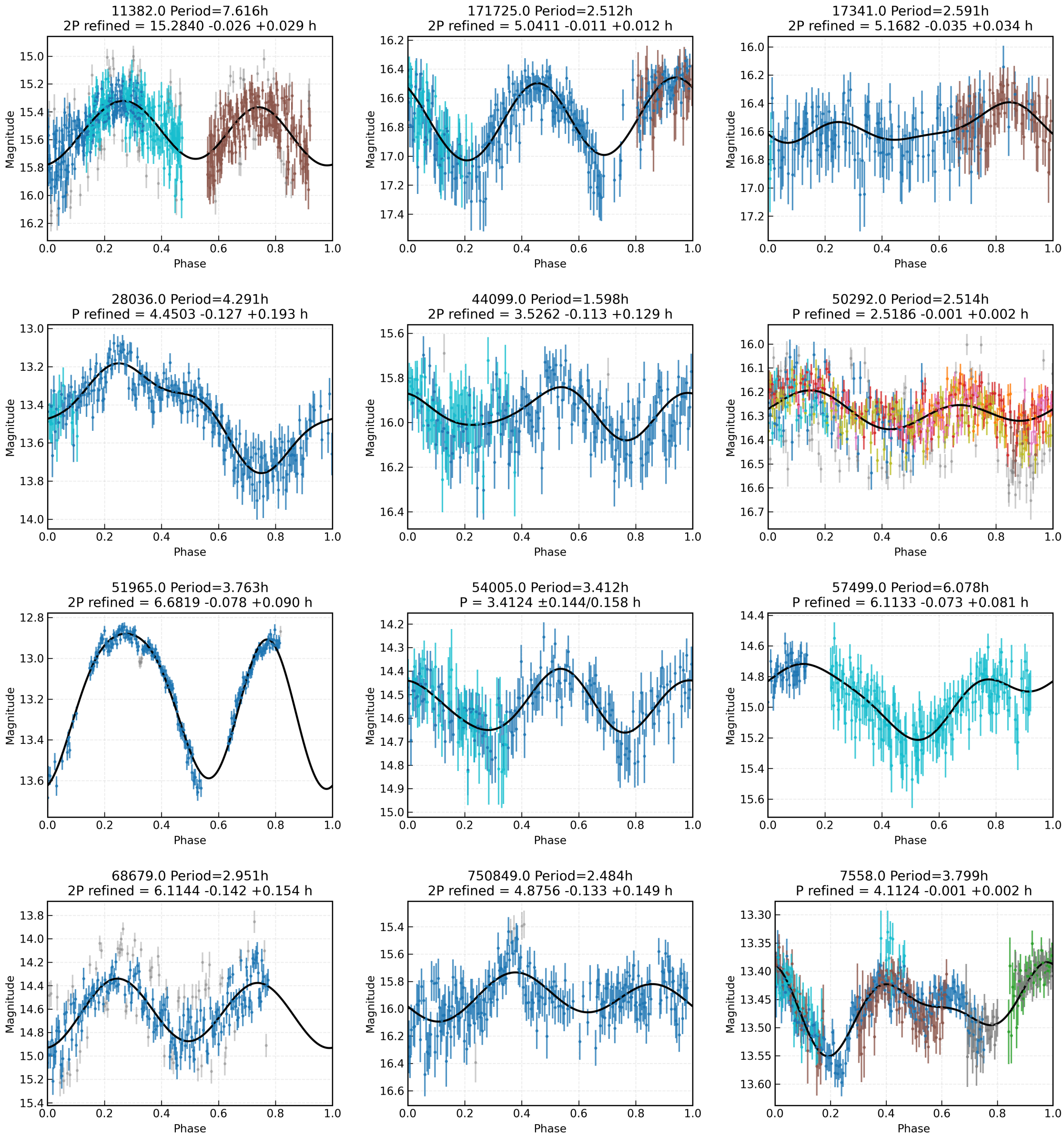

Fig. A.1: Phase-folded light curves for the 12 objects assigned reliable final quality flags in the updated combined GOTTA Prototype and 60 cm Schmidt validation sample. Colored points denote retained measurements from different data subsets, gray points mark rejected or lower-weighted measurements, and black curves show the adopted Fourier models. Each panel gives the object identifier and the candidate period information used in the current period analysis.

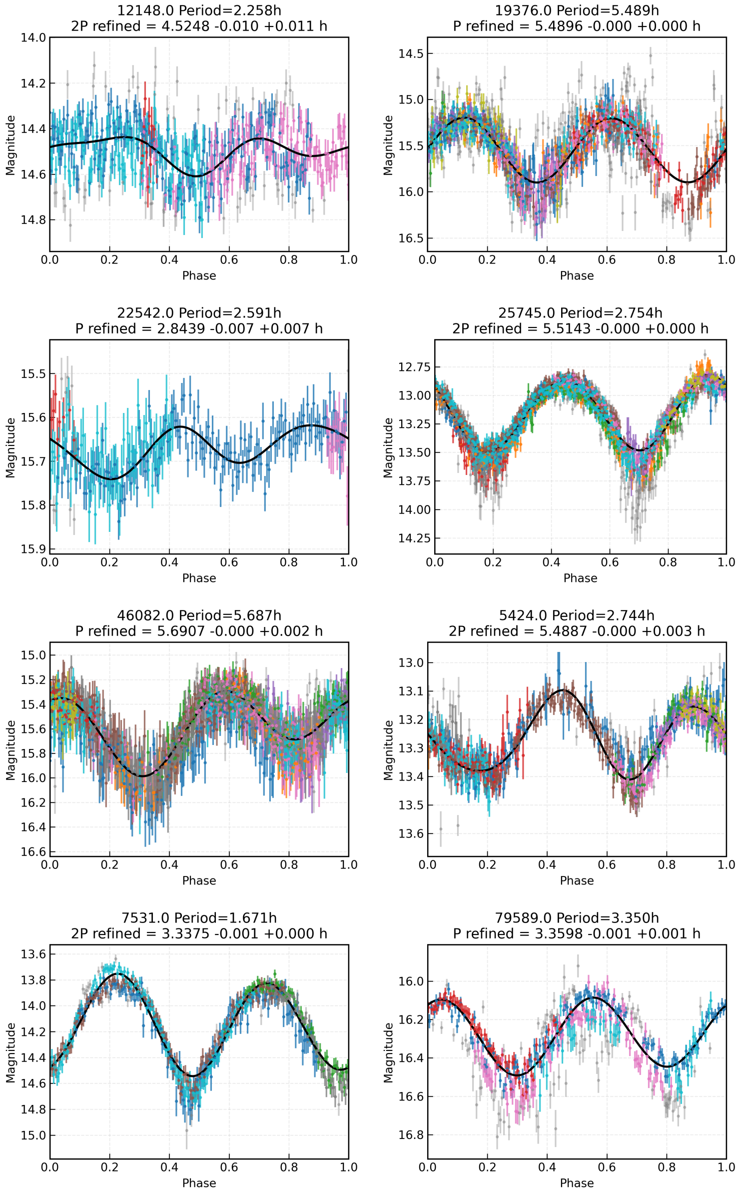

Fig. A.2: Same as Figure A.1, but for the 12 objects assigned questionable final quality flags. These solutions are retained in the validation sample but require additional observations for confirmation.

Table A.1: Final period-analysis table for the combined GOTTA Prototype and 60 cm Schmidt validation sample. N_{total} is the total point count reported in the updated period table, and N_{eff} is the effective number of photometric points retained by the period-analysis quality cuts. The search-quality flag records the agreement among the Lomb–Scargle, PDM, and FFT period searches as summarized by the match-point metric, while the final-quality flag is assigned after the folded-light-curve sampling and period-span assessment. Objects are sorted first by final quality and then by object identifier. Rel., Quest., and Poss. denote reliable, questionable, and possible, respectively.

Object ID	Name	H (mag)	N_{total}	N_{eff}	Span (hr)	P_{rot} (hr)	ΔP (hr)	Model	Phot.	MP	Search qual.	Final qual.
2175	Andrea Doria	14.42	796	671	123.4	4.8880	0.0010	P	aper.	4	Rel.	Rel.
4479	Charlieparker	13.26	1462	1454	195.9	3.0055	0.0197	P	aper.	7	Rel.	Rel.
5424	Covington	13.57	757	722	99.9	5.4887	0.0014	$2P$	Kron	3	Good	Rel.
7531	Pecorelli	14.07	503	503	49.0	3.3375	0.0004	$2P$	PSF	6	Rel.	Rel.
12148	Caravaggio	14.50	459	457	29.5	4.5248	0.0104	$2P$	aper.	5	Rel.	Rel.
19376	1998 BE1	15.18	1093	1066	196.5	5.4896	0.0001	P	aper.	6	Rel.	Rel.
22542	Pendri	15.97	229	190	23.9	2.8439	0.0070	P	Kron	2	Poss.	Rel.
25745	Schimmelpenninck	13.15	1541	1540	170.8	5.5143	0.0000	$2P$	PSF	6	Rel.	Rel.
46082	2001 EC10	15.41	1211	1181	171.6	5.6907	0.0013	P	PSF	6	Rel.	Rel.
72918	2001 RB134	16.66	1473	1466	221.9	6.1385	0.0004	P	PSF	6	Rel.	Rel.
79589	1998 RF19	16.25	443	443	50.2	3.3634	0.0098	P	aper.	9	Rel.	Rel.
97776	2000 KY10	15.31	588	587	78.1	3.5371	0.0417	$2P$	PSF	6	Rel.	Rel.
7558	Yurlov	13.45	438	392	51.3	4.1124	0.0019	P	PSF	–	Rel.	Quest.
11382	Juliasitu	15.46	615	591	53.1	15.2840	0.0273	$2P$	aper.	6	Rel.	Quest.
17341	4120 T-2	16.58	191	189	25.9	5.1682	0.0340	$2P$	PSF	4	Rel.	Quest.
28036	1998 FZ26	13.42	226	222	4.9	4.4503	0.1598	P	PSF	4	Rel.	Quest.
44099	1998 FT117	16.02	216	209	4.8	3.5262	0.1210	$2P$	PSF	6	Rel.	Quest.
50292	2000 CW29	16.29	522	496	53.4	2.5186	0.0015	P	aper.	4	Rel.	Quest.
51965	2001 QO281	13.00	243	207	5.5	6.6819	0.0837	P	PSF	6	Rel.	Quest.
54005	2000 GJ93	14.54	810	760	124.6	3.5362	0.0936	P	PSF	4	Rel.	Quest.
57499	2001 SX239	14.83	206	204	24.0	6.1133	0.0771	P	PSF	4	Rel.	Quest.
68679	2002 CP134	14.56	244	243	4.7	6.1144	0.1479	$2P$	aper.	4	Rel.	Quest.
171725	2000 WW15	16.69	286	276	26.3	5.0411	0.0118	$2P$	PSF	5	Rel.	Quest.
750849	2014 YP35	15.94	215	210	4.8	4.8756	0.1409	$2P$	Kron	8	Rel.	Quest.

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